

SCITECH GROUP · USC ISI

AI Workflows on Laguna using Pegasus

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SciTech Activities

1. Develop technologies to support scientific workflows
2. Provide user support on national cyberinfrastructures and clouds
3. Research new avenues in the workflow management space
4. Provide advice on the design, implementation and use of Cyberinfrastructure (CI) for the Research Infrastructure lifecycle
5. Create awareness and understanding of opportunities in CI for Science

Most projects are collaborative across 4-6 organizations

USC/ ISI



Marina del Rey, CA



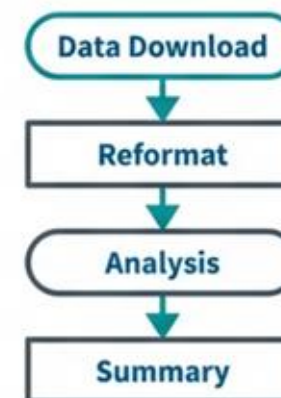
The Challenge: Moving Beyond the Laptop

The Scripting Trap



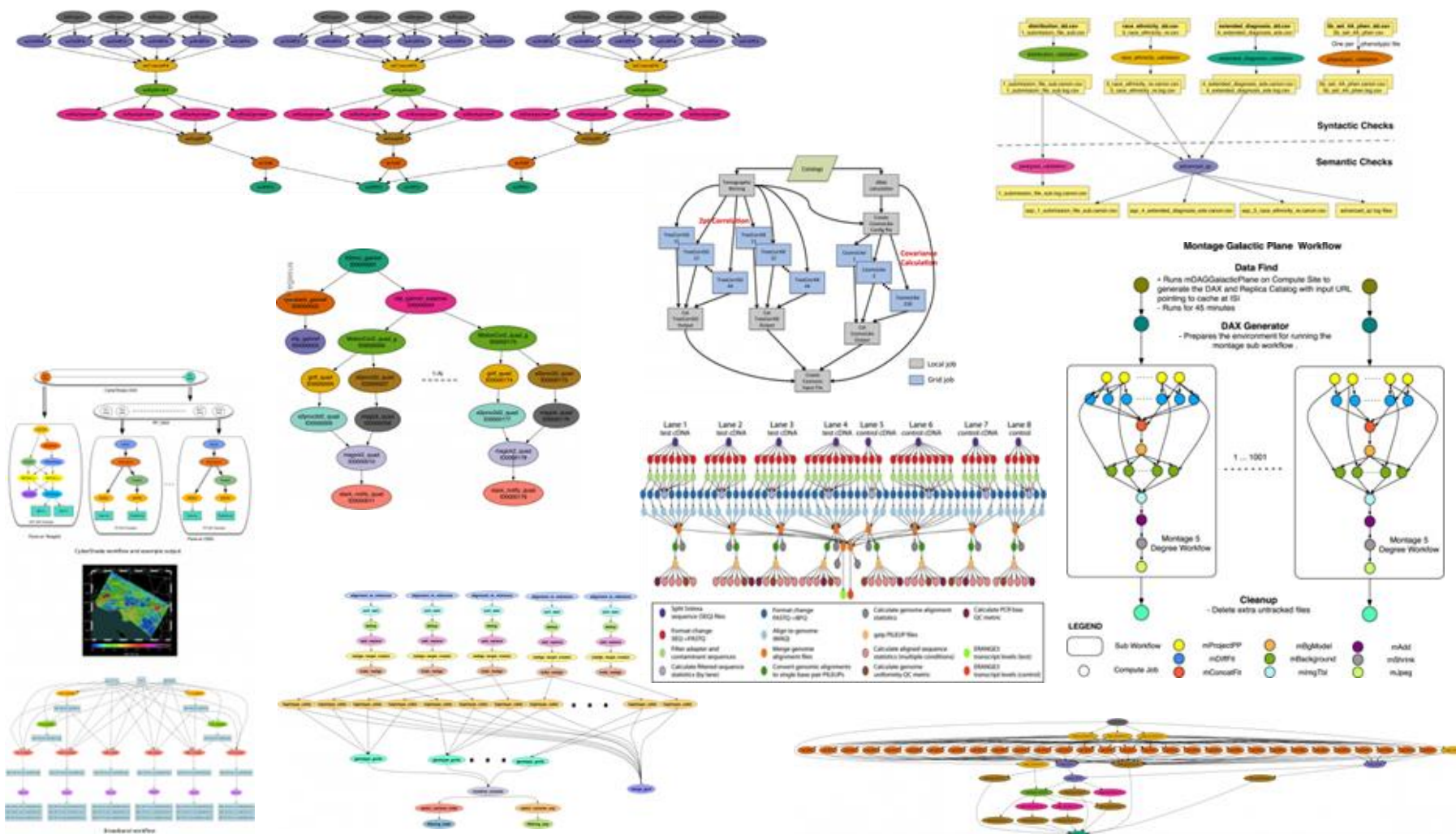
- Quick to code, but hard to scale
- Difficult to maintain and error-prone
- Hard to move between resources (Laptop vs. Cluster)
- “Works on my machine” syndrome

The Workflow Advantage



- Declarative: Describe what to do, not how
- Automated: Minimizes human involvement
- Provenance: Tracks history for reproducibility
- Portable: Runs on ACCESS, NAIRR, cloud, ...

Workflows Built with Pegasus Across Science Domains



Pegasus Workflow Management System, est. 2001

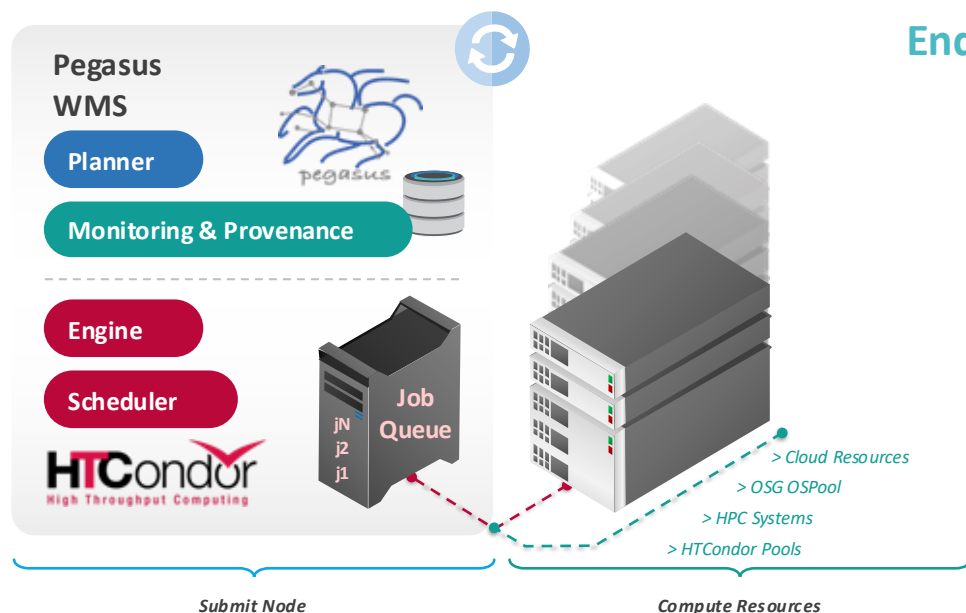
Workflow Challenges Across Domains

- Describe complex workflows in a simple way
- Access distributed, heterogeneous data and resources (heterogeneous interfaces)
- Deal with resources/software that change over time
- Ease of use. Ability to monitor and debug large workflows

Our Focus

- Separation between workflow description and workflow execution
- Workflow planning and scheduling (scalability, performance)
- Task execution (monitoring, fault tolerance, debugging, web dashboard)
- Workflow optimization, restructuring for performance and fault tolerance.

Pegasus Workflow Management System



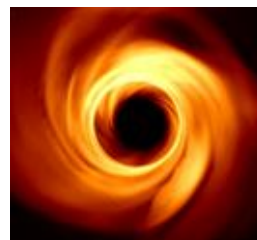
End to End Workflow Management & Execution

- ▶ Develop portable scientific workflows in Python and Java
- ▶ Compile workflows to be run on heterogeneous resources
- ▶ Monitor and debug workflow execution via CLI and web-based tools
- ▶ Recover from failures with built-in fault tolerance mechanisms
- ▶ Regular release schedule incorporating latest research and development

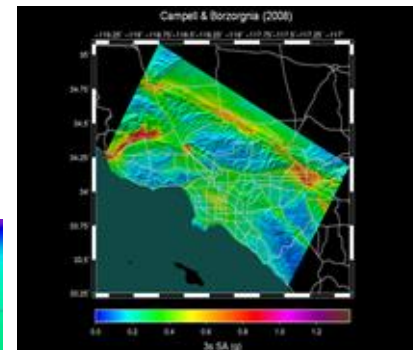
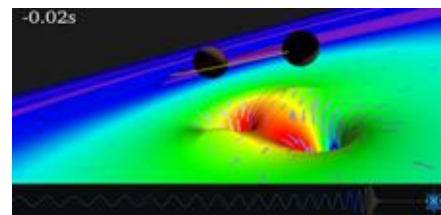
Pegasus in practice

- ▶ Laser Interferometer Gravitational Wave Observatory (LIGO) develops large scale analysis pipelines used for gravitational wave detection.
- ▶ Southern California Earthquake Center (SCEC) CyberShake project generates hazard maps using hierarchical workflows.
- ▶ The Event Horizon Telescope uses Pegasus to orchestrate and scale complex distributed computing pipelines for data calibration, massive synthetic data generation, and deep learning inference

Event Horizon Telescope



LIGO observation of colliding black holes



Hazard map indicating maximum amount of shaking at a particular geographic location generated from SCEC's CyberShake Pegasus workflow

Application Showcase: <https://pegasus.isi.edu/application-showcase/>

Key Pegasus Concepts

▲ Pegasus WMS == Pegasus planner (mapper) + DAGMan workflow engine + HTCondor scheduler/broker

- Pegasus maps workflows to infrastructure
- DAGMan manages dependencies and reliability
- HTCondor is used as a broker to interface with different schedulers

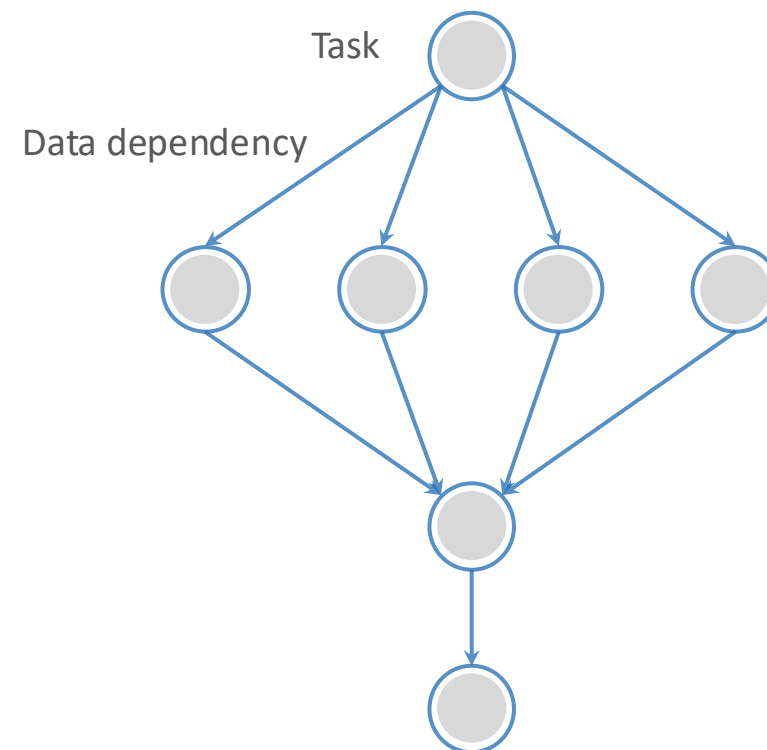
▲ Workflows are DAGs

- Nodes: jobs, edges: dependencies
- No while loops, no conditional branches
- Jobs are standalone executables

▲ Planning occurs ahead of execution

▲ Planning converts an abstract workflow into a concrete, executable workflow

- Planner is like a compiler



Defining Workflows in Python

```
from Pegasus.api import *

# Create abstract workflow
wf = Workflow("pipeline")

# Track the data
page = File("page.html")

# Create a Job
curl_job = Job("curl")\
    .add_args("-o", page, "https://pegasus.isi.edu")\
    .add_outputs(page)

# Create a Dependent Job
wc_job = Job("wc")\
    .add_args("-l", page)\
    .add_inputs(page)

# Add jobs and dependencies
wf.add_jobs(curl_job, wc_job)
wf.add_dependency(wc_job, parents=[curl_job])

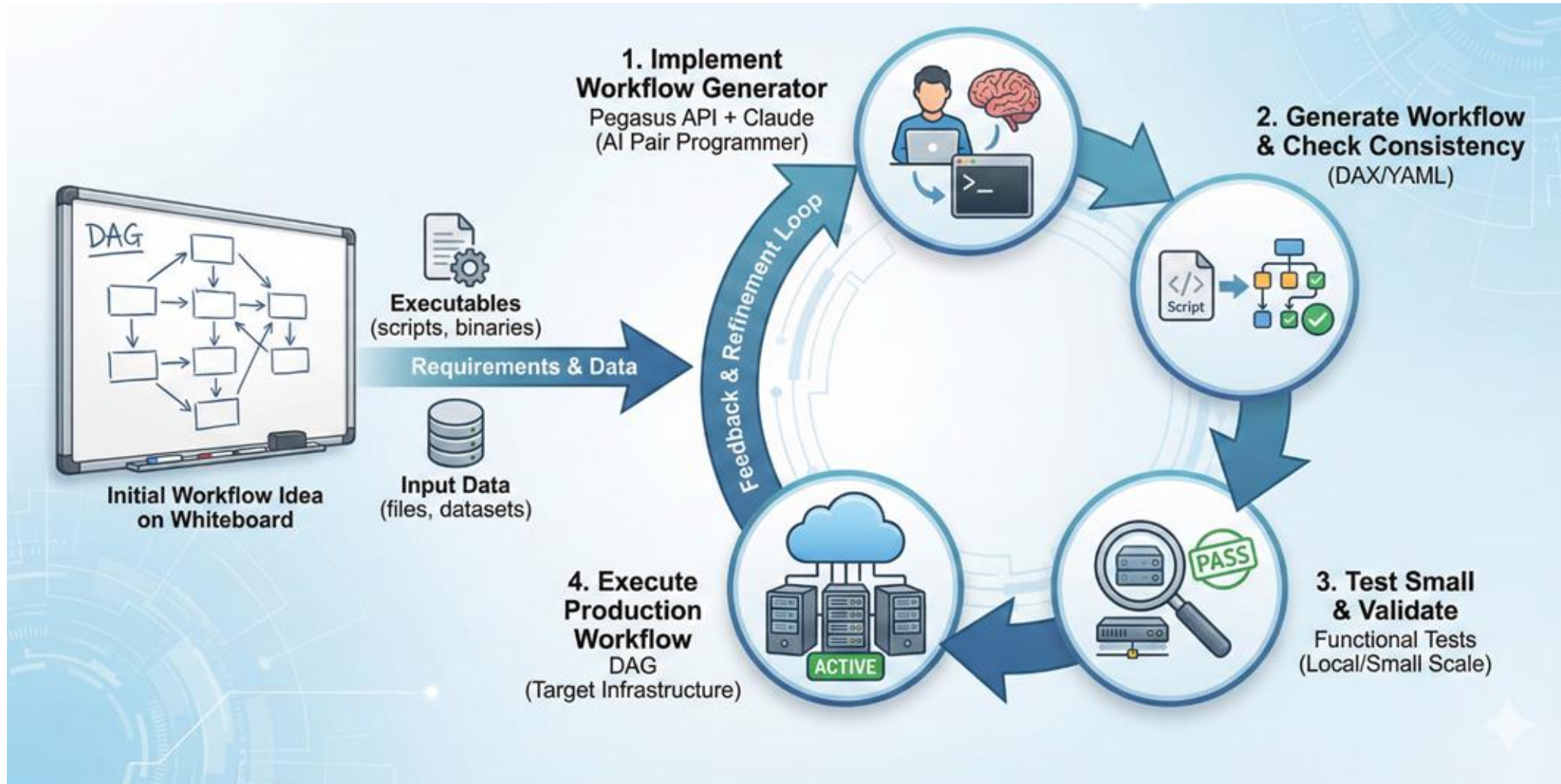
wf.write("pipeline.yml")
```

Declarative job addition

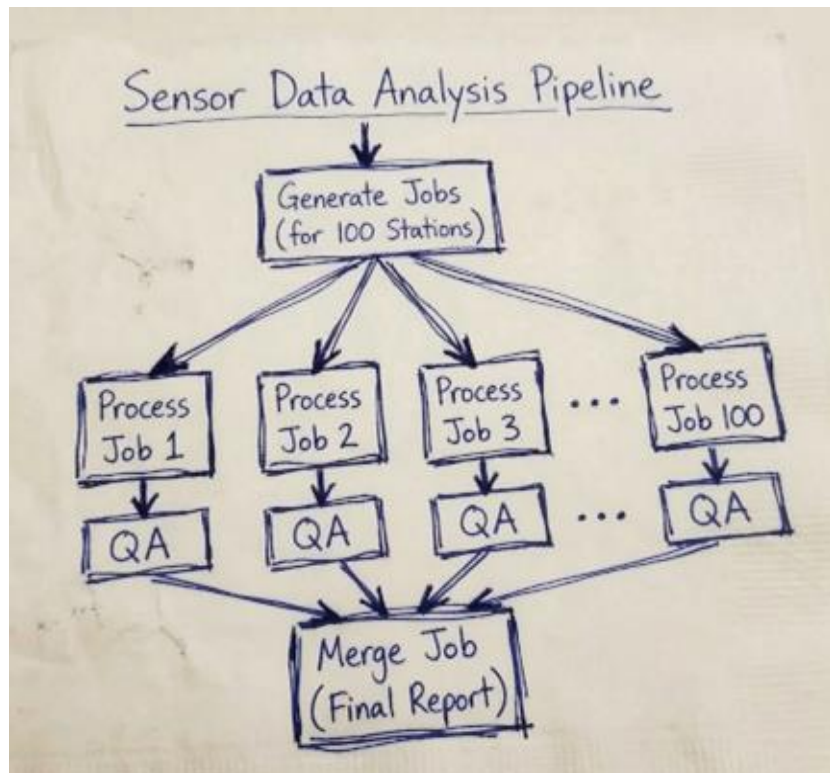
Simple dependency
definition

Generates the abstract
workflow (DAG)

Designing Pegasus WMS Workflows with Claude



From Napkin Sketch to Executable Workflow



Claude reads the image directly

1. Vision analysis

Identifies 4 processing stages, decision diamonds, fork/join points, data-flow arrows

2. Topology extraction

Maps fan-out/fan-in DAG: generate → 100 parallel process+QA branches → merge report

3. Code generation

Produces workflow_generator.py with Pegasus API calls, infer_dependencies=True

4. Script scaffolding

Creates 4 standalone bin/ scripts — generate, process, QA, merge for 100 stations

5. Container + tests

Generates Docker config, test_workflow.py, and parameterized station count

No manual transcription needed. The AI reads the visual spec the way a human collaborator would.

Domain scientists draw what they know — AI translates it into executable workflow code

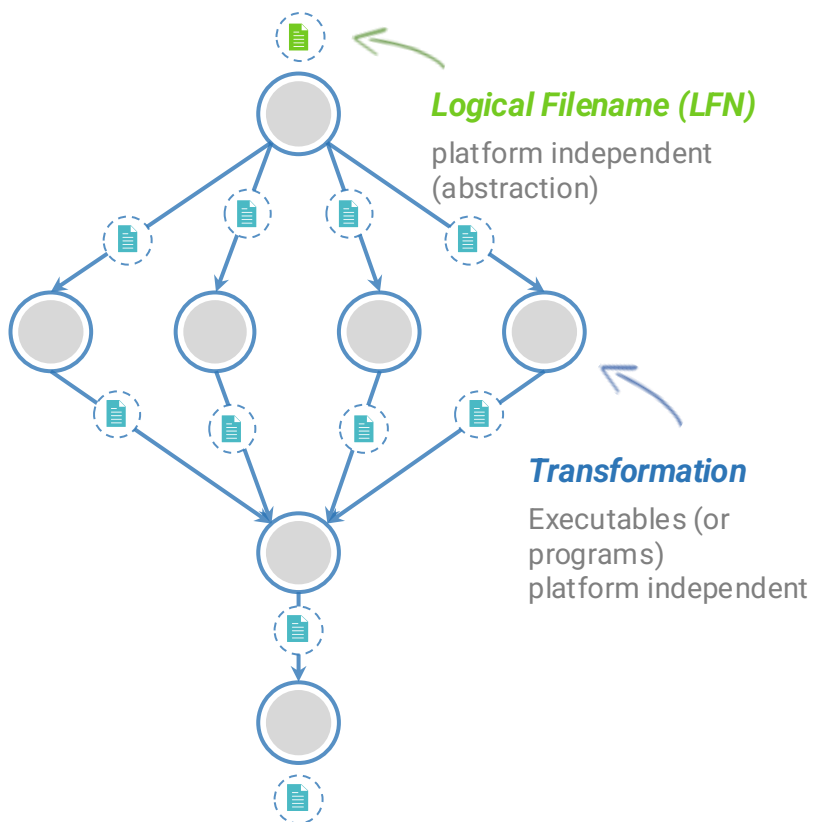
**More on this in the
poster session on
Pegasus AI Studio**

INPUT WORKFLOW SPECIFICATION

Portable Description

Users do not worry about low level execution details

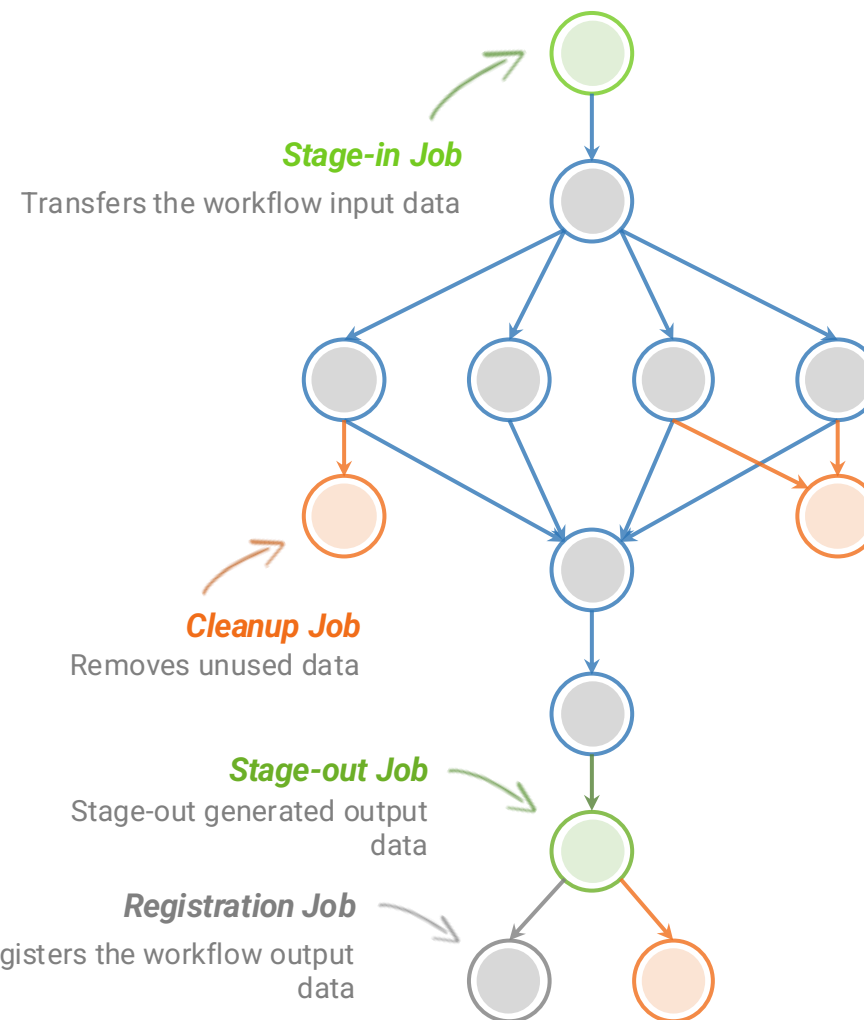
ABSTRACT WORKFLOW



YAML
formatted

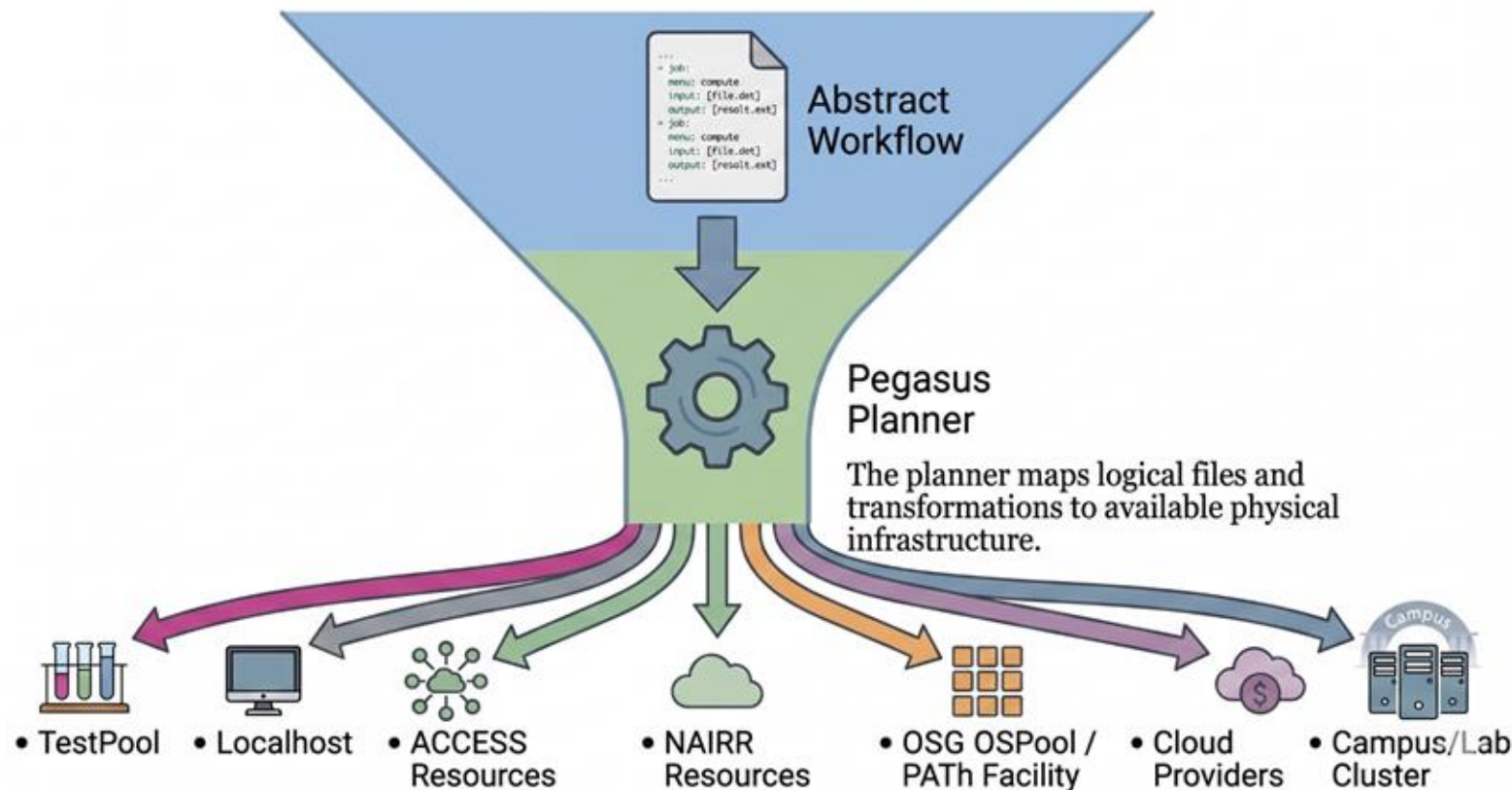


OUTPUT WORKFLOW



EXECUTABLE WORKFLOW

Write Once, Run Anywhere



Requirement: Any HPC/cloud/... system with outbound network connectivity.
Includes your own campus cluster or lab machines!

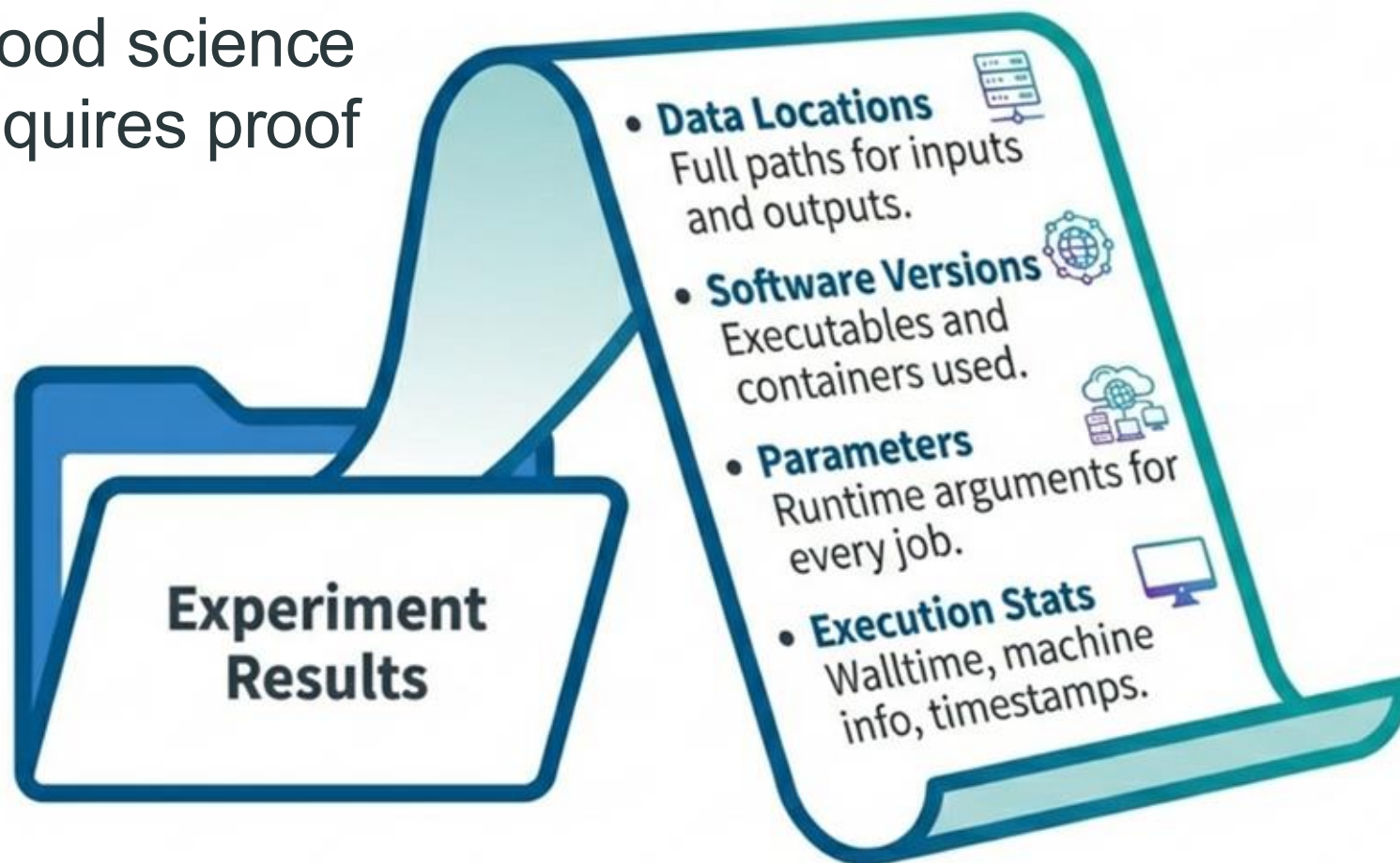
Resilience: When Things Go Wrong



14

Provenance & Reproducibility

Good science
requires proof



Pegasus tracks every bit of **data and every compute** step, ensuring your results are **verifiable** and **reproducible**.

Try Pegasus?

An **easy and convenient** way to try out Pegasus and submit workflows to Laguna is to use the [Laguna Open OnDemand Instance](#).

Hosted, web-based

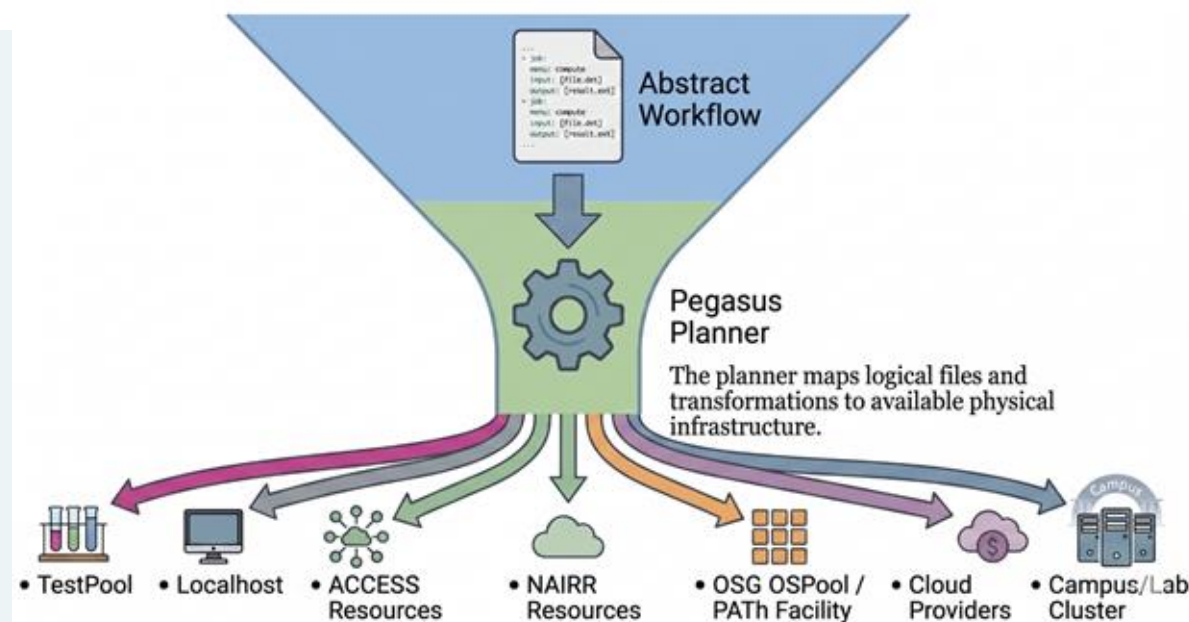
Open OnDemand and Jupyter, with no local install.

Single sign-on with your University Credentials

Training Notebooks

Try out Pegasus training examples

<https://github.com/pegasus-isi/ACCESS-Pegasus-Examples>



Log in: pegasus.access-ci.org

Help: support.access-ci.org/tools/pegasus

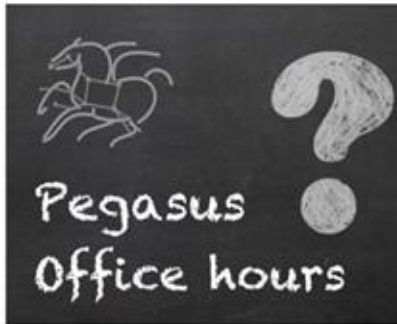
We are here to help!

- SLACK CHANNEL
- EMAIL: PEGASUS-SUPPORT@ISI.EDU

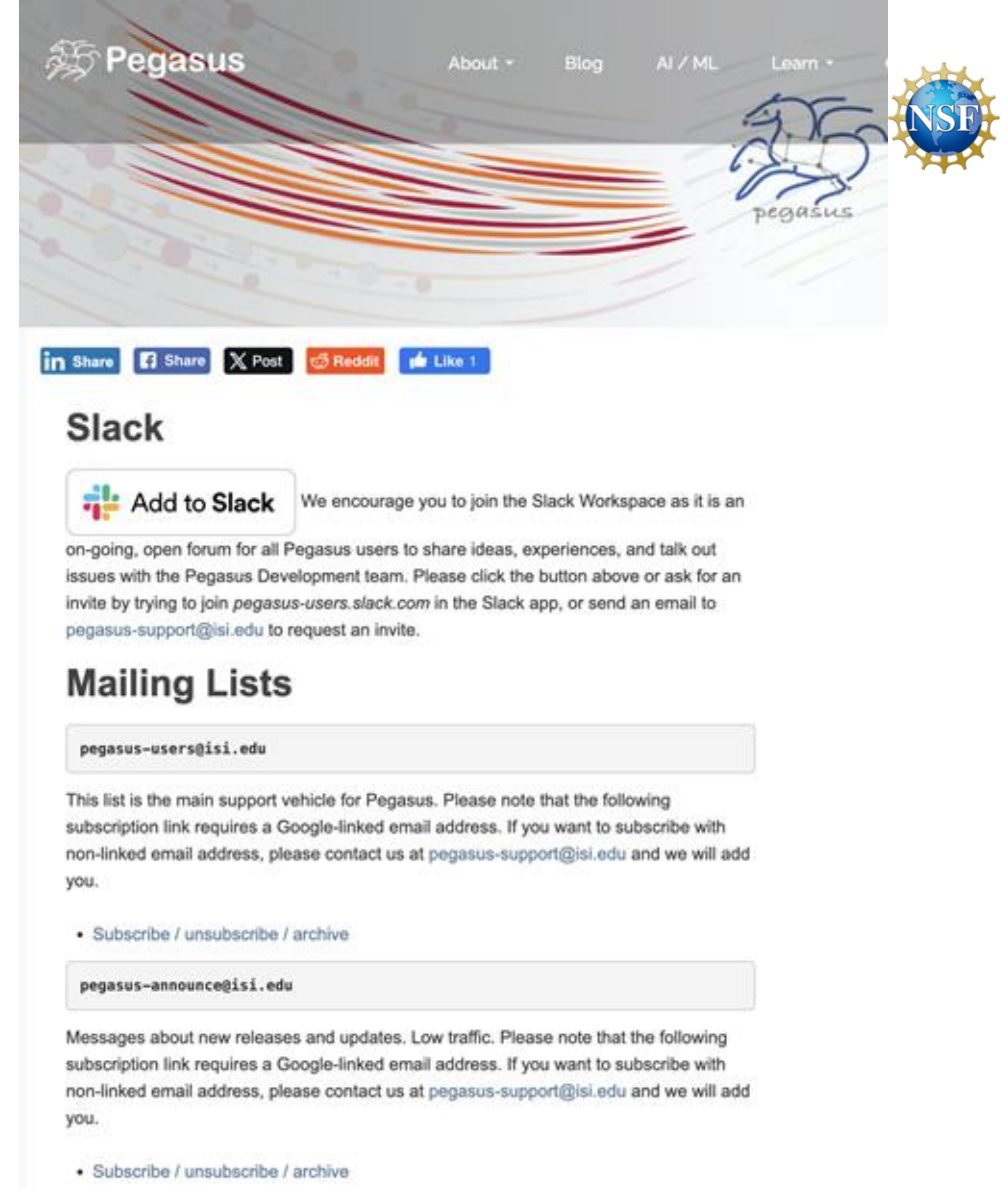
- Office Hours

Join the Pegasus team every Friday for virtual office hours at 11 AM Pacific / 2 PM Eastern.

Do you have questions about workflows or need guidance on organizing and implementing them? Join our weekly office hours – designed to support both new and experienced users in learning and engaging with Pegasus. Here's what to expect:



- TUTORIAL WALKTHROUGH FIRST FRIDAY OF THE MONTH
- [HTTPS://PEGASUS.ISI.EDU](https://pegasus.isi.edu)

A screenshot of the Pegasus website. The header features the Pegasus logo, navigation links for "About", "Blog", "AI / ML", and "Learn", and an NSF logo. The main content area includes social media sharing buttons (LinkedIn, Facebook, X, Reddit, Like), a "Slack" section with an "Add to Slack" button and a description of the Slack workspace, and a "Mailing Lists" section with two email addresses: "pegasus-users@isi.edu" and "pegasus-announce@isi.edu", each with a description of the list and a "Subscribe / unsubscribe / archive" link.

Pegasus About · Blog AI / ML Learn

in Share f Share X Post Reddit Like 1

Slack

Add to Slack We encourage you to join the Slack Workspace as it is an on-going, open forum for all Pegasus users to share ideas, experiences, and talk out issues with the Pegasus Development team. Please click the button above or ask for an invite by trying to join pegasus-users.slack.com in the Slack app, or send an email to pegasus-support@isi.edu to request an invite.

Mailing Lists

pegasus-users@isi.edu

This list is the main support vehicle for Pegasus. Please note that the following subscription link requires a Google-linked email address. If you want to subscribe with non-linked email address, please contact us at pegasus-support@isi.edu and we will add you.

- [Subscribe](#) / [unsubscribe](#) / [archive](#)

pegasus-announce@isi.edu

Messages about new releases and updates. Low traffic. Please note that the following subscription link requires a Google-linked email address. If you want to subscribe with non-linked email address, please contact us at pegasus-support@isi.edu and we will add you.

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Success Stories

Pegasus for Seismic Hazard Applications

45

Days

Duration of 24/7 operation

55K

Frontier

Node-hours

125K

Frontera

Node-hours

GPU

CPU

*High-water marks were 4130 nodes on Frontier (44%) and 1029 nodes on Frontera (12%)**

1 PB

Data

Managed by Pegasus

9M

Files

Staged to USC CARC

945

Workflows

27,720 jobs

**Useful information
for:**

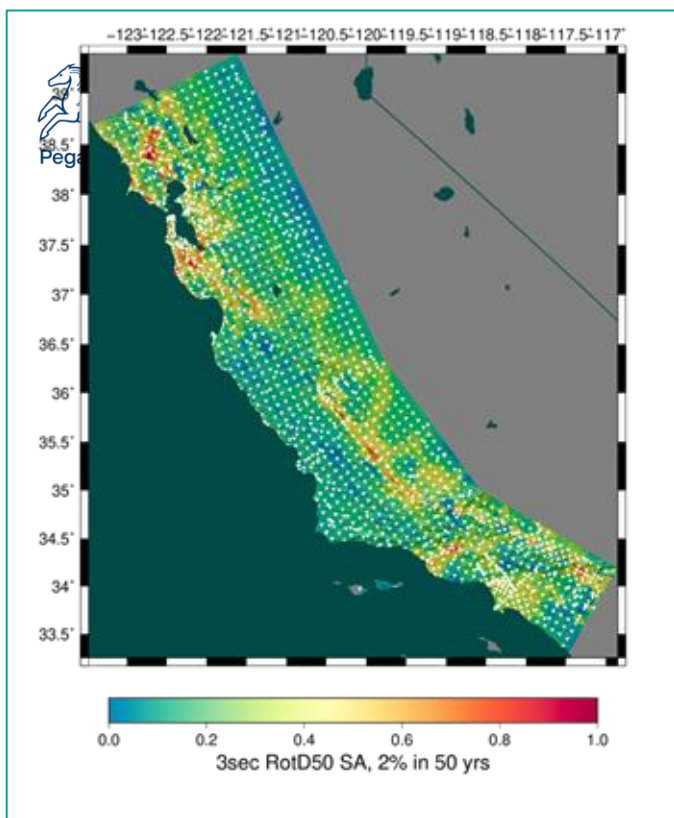
- ✓ Building engineers
- ✓ Disaster planners
- ✓ Insurance agencies

Two new Northern California hazard models were produced.



Slide Courtesy of Scott Callaghan, USC

* Neither used a reservation or priority bump



XENONnT - Dark Matter Search



Two Workflows

Monte Carlo simulations and the main processing pipeline.

- Workflows execute across Open Science Grid (OSG) & European Grid Infrastructure (EGI)
- Rucio for data management
- MongoDB instance to track science runs and data products.



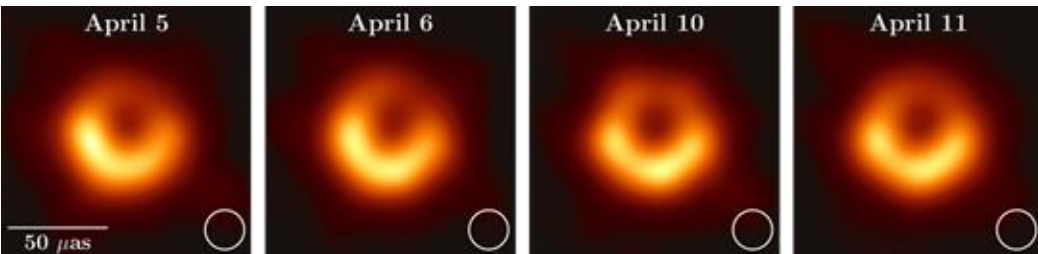
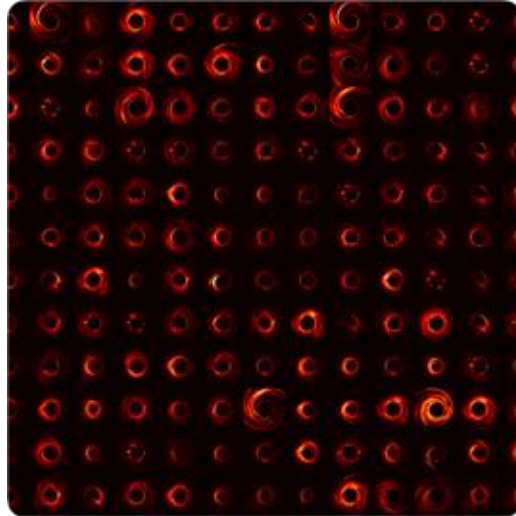
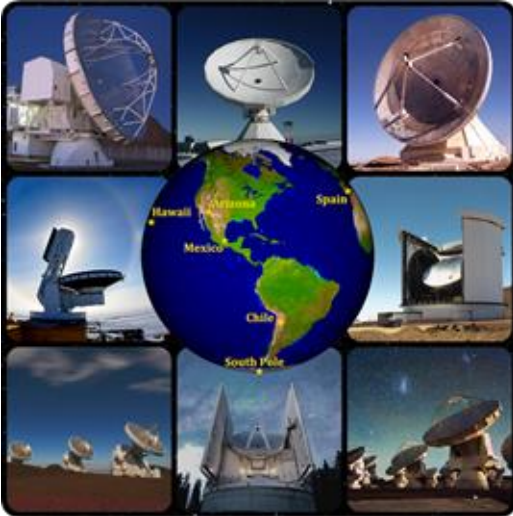
Type	Succeeded	Failed	Incomplete	Total	Retries	Total+Retries
Tasks	4000	0	0	4000	267	4267
Jobs	4484	0	0	4484	267	4751
Sub-Workflows	0	0	0	0	0	0

Workflow wall time : 5 hrs, 2 mins
 Cumulative job wall time : 136 days, 9 hrs
 Cumulative job wall time as seen from submit side : 141 days, 16 hrs
 Cumulative job badput wall time : 1 day, 2 hrs
 Cumulative job badput wall time as seen from submit side : 4 days, 20 hrs

Event Horizon Telescope Bringing Black Holes into Focus

8 telescopes: 5 PB of data

60 simulations: 35 TB data



First images of black hole at the center of the M87 galaxy

Improve constraints on Einstein's theory of general relativity by 500x

480,000 jobs - 2,600,000 core hours

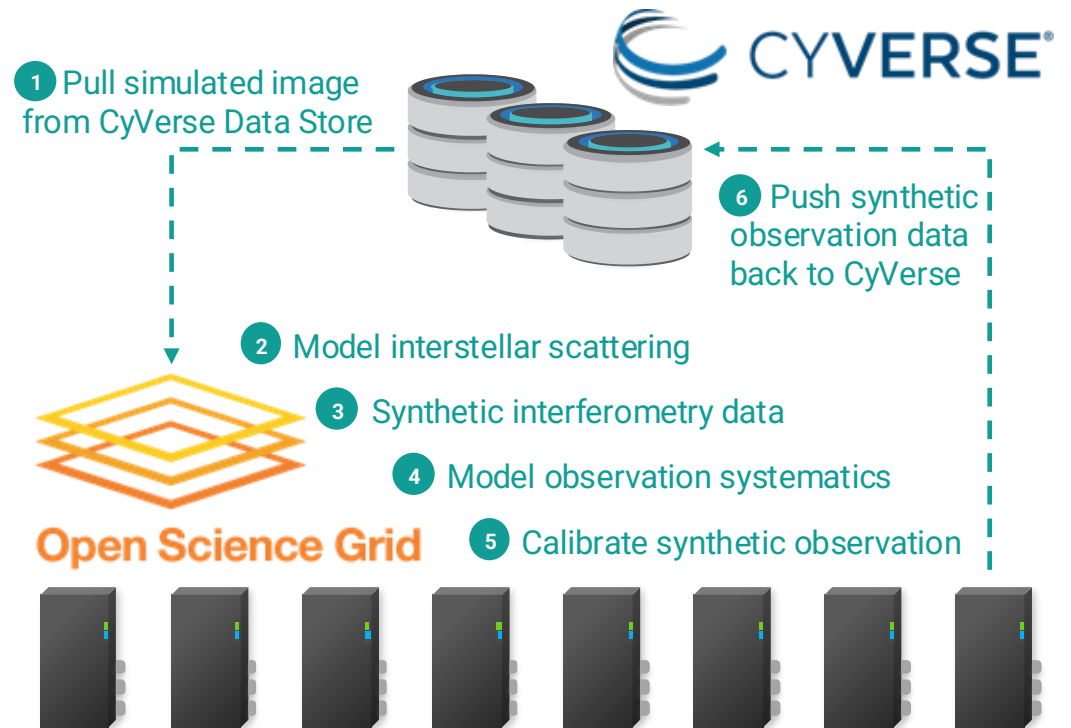
#15 in all OSG projects in last 6 months

#2 in all OSG astronomy projects in the last 6 months

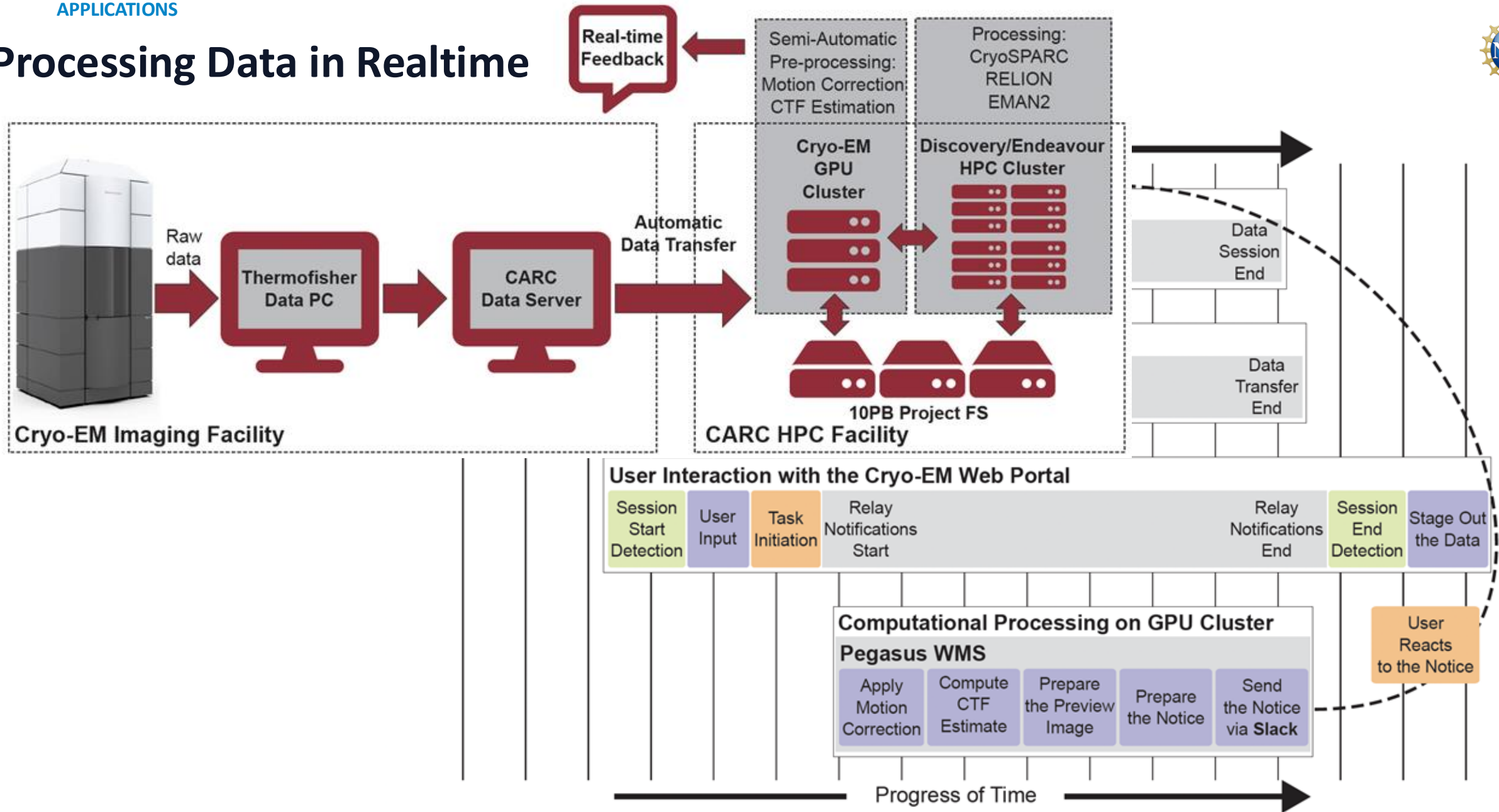


Pegasus-SYMBA Pipeline

Physically accurate synthetic observation data from simulations are keys to develop calibration and imaging algorithms, as well as comparing the observation with theory and interpreting the results.



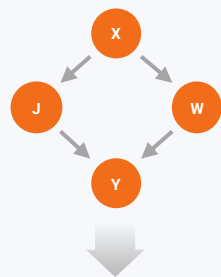
Processing Data in Realtime



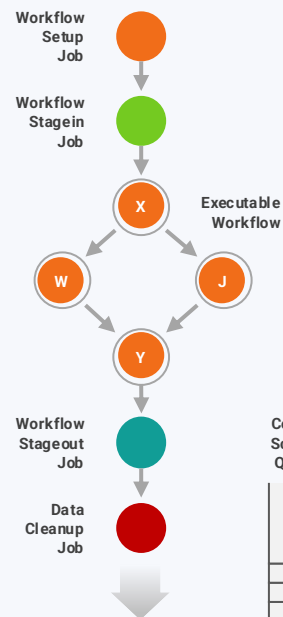


SUBMIT HOST

Abstract Workflows



Pegasus Planner



Condor DAGMan

Input data hosted at LIGO Sites



Nebraska GridFTP Data staging server
GridFTP, HTTP, SRM



Input Files



Intermediate Files



Produced Dataset

LIGO output data server



1

Workflow stagein Job stages in the input data for workflow from user server

2

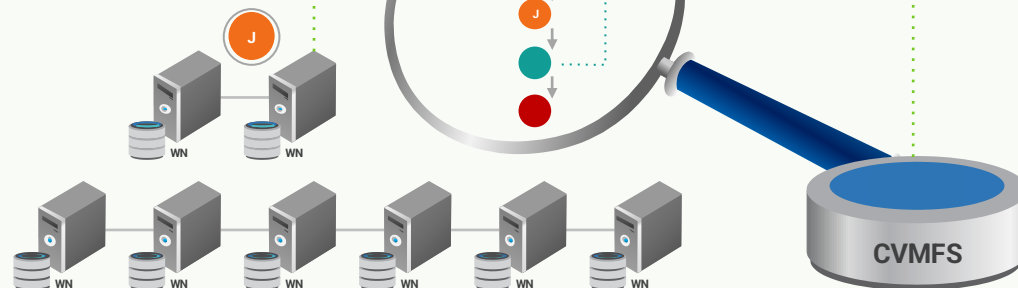
PegasusLife instance looks up input data on the compute node/ CVMFS If not present, stage-in data from remote data staging server

4

Workflow Stageout Job stages produced data from data staging server to LIGO Output Data Server

HTTP Squid Cache

Nodes from OSG & LIGO Sites managed By Glidein WMS



3

Pegasus lite instance stages out job output data from worker node to data staging server

LEGEND



Directory Setup Job



Data Stagein Job



Data Stageout Job



Directory Cleanup Job



Pegasus Lite Compute Job



Worker Node



PEGASUS WORKFLOWS Advanced LIGO

60,000 Compute Tasks

Input Data: 5000 files (10GB total)

Output Data: 60,000 files (60GB total)

Processed Data: 725 GB

Laser Interferometer
Gravitational Wave Observatory



First Detection Runs executed on
LIGO Data Grid, EGI, Open Science
Grid and XSEDE

Thank you! Questions?

Karan Vahi | vahi@isi.edu

- **Acknowledgments**

For support on making Pegasus available at Laguna

TO

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BH

Bret Harry

Systems Administrator. ITS. USC.

JH

James Hong

Cold Front Developer. CARC. USC